MICHAEL VASQUEZ

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U.S. Citizen • Clearance Eligible

PROFESSIONAL SUMMARY

Mechanical Engineer with full-lifecycle product development experience across aerospace lighting, life-sciences automation, and precision mechanism design. Demonstrated proficiency in requirements-driven design, FEA-based structural optimization, model-based simulation, and DFM under regulated standards (FAA, UL, NEMA, ISO 9001). Skilled at integrating mechanical hardware with controls and software to deliver compliant, certifiable systems on schedule.

CORE COMPETENCIES

Design & Analysis: Mechanical Design, Precision Mechanisms, GD&T (ASME Y14.5), Tolerance Analysis, FEA (Static, Modal, Impact), DFM/DFA, Requirements Analysis, Verification & Validation (V&V)

CAD/CAE Tools: SolidWorks, Creo, Fusion 360, AutoCAD, OnShape, ANSYS

Modeling, Simulation & Programming: MATLAB, Simulink, Python, C++, Java, System Dynamics, Flight Dynamics Modeling, Numerical Integration

Manufacturing & Metrology: CNC (Mill/Lathe), Additive Manufacturing (FDM/SLA), CMM, ShapeGrabber Ai620, Ultrasonic Welding, Injection Molding

Standards & Compliance: FAA Obstruction Lighting, UL 50E, NEMA 4X, ISO 9001

Certifications: GD&T Professional, CSWA Mechanical Design, CSWA Additive Manufacturing

PROFESSIONAL EXPERIENCE

Mechanical Engineer — Point Lighting Corporation, Bloomfield, CT June 2025 – Present

- Lead full product lifecycle engineering — concept, requirements, CAD, prototyping, qualification testing, and certification — for aerospace obstruction lighting under ISO 9001; coordinate cross-discipline reviews with EE and SWE
- Designed an injection-molded junction box to replace cast aluminum predecessor; reduced part count, simplified supplier base, and cut projected unit cost while maintaining UL 50E and NEMA 4X compliance
- Designed a solar-powered mounting fixture for FAA L-810/L-864 obstruction lighting that streamlined field assembly and reduced bill-of-materials cost vs. legacy hardware
- Restructured rapid prototyping workflow, reducing prototyping cost by 90% while preserving FAA-compliant test fidelity

Manufacturing Engineering Intern — Thermo Fisher Scientific, Penfield, NY Jan – Aug 2023, Jan – Aug 2024

- Designed and deployed automated part-presence detection systems for production lines, increasing first-pass manufacturing yield by 60%
- Reconfigured PLC controls and Epson SCARA robot end-effector sequencing on capping cell, increasing part yield by 18%
- Engineered automated CMM and ShapeGrabber Ai620 measurement fixturing to reduce dimensional inspection cycle time and accelerate production throughput
- Standardized cutting/reaming and ultrasonic welding process parameters, achieving 98% process yield across multiple SKUs

ENGINEERING PROJECTS

Binder-Jet Bone 3D Printer — Senior Capstone (Client: RevBio) Aug 2024 – Apr 2025

- Scaled an open-source binder-jet additive manufacturing platform up 75% on an 8-engineer team to print bone scaffolds using RevBio Tetranite bioactive bone adhesive
- Designed a 2-DoF compliant coupling for the Z-axis piston that eliminated the requirement for precision-machined linear bearing mounts and resolved tolerance stack-up impeding linear motion
- Redesigned piston outer sleeve as a 3D-printed split-half assembly enabling shim-based fine-tuning of powder seal compression, reducing tooling cost while meeting precision requirements

Custom FPV Drone — Frame Design, Flight Dynamics Modeling, FEA Dec 2024 – Present

- Built a custom MATLAB flight dynamics simulator from first principles (Newton's Second Law, Euler integration) modeling thrust, drag, gravity, and mass; validated outputs against industry flight calculators and used derived terminal velocity and impact loads as load cases for downstream FEA
- Performed FEA-driven arm optimization in ANSYS for side, bending, and max-acceleration load cases; achieved factor of safety of 2 with sacrificial geometry engineered to fracture outboard of the central frame to protect avionics

Three-Stage Helical Planetary Gearbox — Mechatronic System Jan – Aug 2024

- Designed a 27:1 three-stage planetary gearbox to motorize a manually cranked standing desk; iterated spur to helical teeth to reduce acoustic emission and redesigned planet carrier to resolve v1 torsional failure; sized gears via load testing and dynamic analysis with ~3x factor of safety, integrated with custom motor controller (variable speed, direction, voltage telemetry)

LEADERSHIP

President — Rochester Institute of Technology VEX U Robotics May 2021 – May 2022

- Led 15+ member hardware/software sub-teams through design, fabrication, and integration of two competition robots; team won the Innovate Award at VEX U World Championships and ranked 3rd in qualifications. Developed parallel-PID controller with odometry-based localization for autonomous disturbance rejection

EDUCATION

B.S. Mechanical Engineering — Rochester Institute of Technology Aug 2020 – May 2025